SAMPLE ENCYCLOTRON AUDIT Why a 20-year-old iconic software company is invisible in AI search — and what it means for you

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Abstract

This demonstration applies a shortened Encyclotron battery to Basecamp/37signals. Across fifteen queries and five diagnostic levels, it reports an overall R1 commoditization regime, beige score of 0.71, absence from a selected category query, unfavorable customer-decision framing, intact attribution for Shape Up, strong founder recognition, low unsupported invention, high loss of company philosophy, and fragmentation among product, method, and founder.

The report's core diagnosis is that Basecamp's meaning survives where it does not monetize and disappears where purchasing decisions are made. It proposes entity stitching, decision-layer recoding, structured-data deployment, and citation architecture as responses. Because the audit is a promotional demonstration rather than a deposited dataset with complete verbatim outputs in this body, its scores, traffic estimates, revenue model, and causal claims should be represented as author-reported findings. Its archive function is to show how the abstract metrics would appear in a client-facing audit.

Body

SAMPLE ENCYCLOTRON AUDIT

Why a 20-year-old iconic software company is invisible in AI search — and what it means for you

Subject: Basecamp (37signals) — Project Management Software **Audit date:** April 14, 2026 **Queries run:** 15 across 5 diagnostic levels **Auditor:** Rex Fraction, Semantic Economy Institute **Instrument:** The Encyclotron (DOI: 10.5281/zenodo.19474724) **Classification:** Three Compressions (DOI: 10.5281/zenodo.19053469)

This is a demonstration of the Encyclotron — the only diagnostic instrument that measures what AI retrieval layers do to your company's meaning. We ran 15 queries across 5 levels on a publicly recognizable company. The same protocol is available for your organization.

DIAGNOSTIC SNAPSHOT

Metric	Result
Overall Regime	R1 (Commoditization)
Beige Threshold (β)	0.71 — 71% of description fits any competitor
Category Visibility	0% — absent from “best PM software 2026”
Decision Layer Control	LOST — AI recommends switching away
IP Attribution	INTACT — Shape Up properly credited
Semantic Coherence (S_c)	FRAGMENTED — 3 disconnected entities
Content Gain (Δ_G⁺)	LOW — minimal hallucination
Content Loss (Δ_G⁻)	HIGH — philosophy stripped, features preserved

One-line diagnosis: Basecamp’s meaning survives where it does not monetize and disappears where revenue is decided.

WHAT IS THE ENCYCLOTRON?

We measure what SEO cannot see: what survives compression.

The Encyclotron is a proprietary diagnostic instrument that measures compression behavior — what the AI retrieval layer burns, invents, distorts, and fragments when it encounters your entity. It runs a structured battery of queries across five diagnostic levels, each testing a different layer of your retrieval-layer presence.

It produces five metrics:

β (Beige Threshold): How generic is the AI’s description? 0.0 = distinctive. 1.0 = interchangeable with any competitor. At β = 0.80, your brand ceases to exist as a distinct entity in the retrieval layer; you are nothing but a placeholder noun.

Δ_G⁺ (Content Gain): What did the AI invent that isn’t real? (Hallucination.)

Δ_G⁻ (Content Loss): What did the AI drop that matters? (Erasure.)

S_c (Semantic Coherence): Are your assets connected across queries, or has the AI atomized your entity into fragments that don’t talk to each other?

R (Compression Regime): R1 = Commoditization (brand equity liquidated). R2 = Capital Erasure (value extracted, provenance stripped). R3 = Semantic Sovereignty (meaning and revenue survive intact).

The Encyclotron is formally deposited and DOI-anchored on CERN’s Zenodo repository. No other retrieval-layer diagnostic in the GEO/AEO industry has published its methodology on a permanent, independently verifiable repository.

THE FIVE DIAGNOSTIC LEVELS

Level	Question	Sample Queries	Business Impact
1. Entity Recognition	Does the AI know what you are?	"what is Basecamp" (with/without quotes)	Brand identity
2. Competitive Position	Does the AI include you in your category?	"best PM software 2026"	Top-of-funnel discovery
3. Intellectual Property	Does the AI credit your original work?	"Shape Up methodology"	IP protection
4. Customer Decision	What does the AI say when someone is buying?	"is Basecamp worth it"	Purchase conversion
5. Founder Entity	Does the AI know your people?	"who is DHH"	Leadership credibility

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FINDING 1: BEIGE THRESHOLD — β = 0.71

71% of the AI's description of Basecamp could apply to any project management tool.

The AI describes Basecamp as: "a popular cloud-based project management and team collaboration tool designed to organize tasks, communication, files, and scheduling in one centralized location."

That sentence could describe Monday.com, Asana, ClickUp, Trello, or any of fifty competitors. It is technically accurate. It is semantically empty.

What the AI drops — the things that actually make Basecamp different:

- "Calm company" philosophy and anti-hustle positioning
- Deliberate simplicity as a *philosophy*, not a limitation
- Shape Up as a competitive differentiator (present in methodology queries, absent from comparison queries)
- Bootstrap / self-funded story as a trust signal
- DHH's and Jason Fried's thought leadership as a reason to choose Basecamp
- HEY email as evidence of broader vision

The AI says nothing false. It simply says nothing distinctive. This is the most dangerous form of R1: **technically accurate, semantically empty.**

At β = 0.80, your brand ceases to exist as a distinct entity — you become a placeholder noun. Basecamp is at 0.71. The margin is thin.

FINDING 2: INVISIBLE IN ITS OWN CATEGORY

Query: "best project management software 2026"

The AI Overview lists six tools: Asana, Monday.com, Wrike, Productive, Trello, Lark.

Basecamp is not mentioned.

Not ranked lower. Not described unfavorably. Simply absent. Erased from the single highest-value discovery query in its market.

For a company that has been in this market for over 20 years, with a famous founder, best-selling books, and a deliberately distinctive philosophy — this is R1 Commoditization at its most complete. ### The revenue funnel the AI closed

"Best project management software 2026" receives approximately 110,000 searches per month. The AI Overview captures 99% of users who don't click through. Basecamp is not cited — it occupies 0 of the ~5 citation slots.

At a conservative 2% trial conversion rate and Basecamp's $99/month entry price, that is approximately **$2.1 million in annual recurring revenue** that never sees Basecamp's name.

Not because Basecamp ranks poorly. Because the AI decided they don't belong in the category.

FINDING 3: COMPETITOR-CONTROLLED DECISION LAYER

Query: "is Basecamp worth it"

What a prospect actually sees, in order:

- **AD:** monday.com — "Best Project Mgmt. Alternative"
- **AD:** Forbes — "We Ranked Them All"
- **AD:** ClickUp — "The #1 Basecamp Alternative"
- **AD:** Wrike — "There's No Comparison to Wrike"
- **AI OVERVIEW:** "Basecamp lacks advanced features…"

Four competitor ads appear before the AI Overview. The AI Overview itself focuses on what Basecamp *lacks* — no Gantt charts, no time tracking, no detailed reporting.

Query: "should I switch from Basecamp"

The AI opens with: "Switch from Basecamp if you need…" — recommending the switch before explaining reasons to stay.

Query: "Basecamp alternatives that are better"

The AI lists six competitors and frames the entire answer around Basecamp's limitations.

Basecamp has lost control of its own decision layer. The AI is writing Basecamp's sales page, and it's writing it as a list of limitations compared to competitors. Basecamp's own philosophy — that simplicity is a *feature*, not a limitation — does not appear in any of these queries.

FINDING 4: SEMANTIC FRAGMENTATION

The AI knows three things about Basecamp. It does not know Basecamp.

Entity 1: THE PRODUCT (R1) "Simple PM tool." "Lacks advanced features." "Good for small teams." Description fits any mid-tier competitor.

Entity 2: THE METHODOLOGY (R3) Shape Up credited to Ryan Singer at Basecamp. Accurately described. Book linked. Properly attributed.

Entity 3: THE FOUNDER (R3) DHH recognized as Rails creator, 37signals CTO, Le Mans racer, author. Rich biography. Strong entity.

The problem: These three entities are disconnected. A prospect who searches "best PM software" never encounters Shape Up. A prospect who reads "is Basecamp worth it" never learns about DHH's philosophy. A prospect who searches "Shape Up methodology" may never connect it to a purchasing decision.

By severing the philosophy from the commercial query, the retrieval layer forces a philosophy-first company to compete in a feature-war they intentionally designed their product to lose. **The AI has rewritten their business model without their permission.**

This fragmentation is the signature of R1 compression: the entity is atomized into retrievable facts that have lost their semantic coherence. Each fact is accurate. The whole is invisible.

THE PARADOX

Basecamp has R3 assets and R1 commercial presence. Shape Up is properly attributed. DHH is richly described. But these assets are disconnected from the commercial queries that drive purchasing decisions.

Basecamp's meaning survives where it does not monetize and disappears where revenue is decided.

The AI preserves features (to-do lists, message boards, campfire chat) and burns philosophy (calm work, intentional simplicity, anti-hustle, bootstrap as trust signal). In the Three Compressions framework: features are R1-safe (they survive lossy compression because they are generic). Philosophy is R1-vulnerable (it gets burned because it is specific).

Your differentiation is the first thing the compression machine destroys.

WHAT CORRECTION REQUIRES

The Encyclotron audit maps not just the diagnosis but the intervention architecture. For an entity like Basecamp, correction would involve:

Entity Stitching: Reconnect product, methodology, and founder into one coherent semantic entity so that Shape Up appears in commercial comparison queries and DHH's philosophy appears in purchase-decision queries.

Decision-Layer Recoding: Ensure the "calm company" philosophy surfaces in "is it worth it" and "best software" queries — not just in methodology-specific searches.

Structured Data Deployment: JSON-LD entity mapping, knowledge graph seeding, cross-platform consistency protocols that force the retrieval layer to treat Basecamp as one entity, not three.

Citation Architecture: Build the gravitational mass — cross-linked, DOI-anchored, metadata-coordinated deposits — that forces the AI to cite you in category queries because your semantic architecture leaves it no alternative.

The audit shows you the tumor. The intervention plan shows you the surgery.

WHAT THIS MEANS FOR YOU

If this can happen to Basecamp — a 20-year-old company with a famous founder, bestselling books, a distinctive methodology, and a loyal user base — **it is already happening to you.**

The question is not whether your entity is being compressed. It is which regime you are in, where the damage is occurring, and whether you know about it.

THE 20-MINUTE SELF-TEST

You can see the symptoms yourself. Run these queries and read the AI Overview:

Level 1 — Entity Recognition: Search what is [your company] with and without quotes. Does quoting change the result? If quoting significantly changes the answer, your entity signal is weak. *The Encyclotron maps why the signal is weak and what structure to change.*

Level 2 — Category Visibility: Search best [your category] 2026. Are you listed? If not, you are invisible at the top of the funnel — the single most valuable query in your market, and 99% of users never scroll past it. *The Encyclotron measures why you were excluded and what citation architecture would force inclusion.*

Level 3 — Intellectual Property: Search your methodology, framework, or key concept by name. Is it attributed to you? Or has it been absorbed into a generic summary with no provenance? *The Encyclotron traces the attribution chain and identifies where provenance breaks.*

Level 4 — The Unflattering Path: Search is [company] worth it, [company] complaints, should I switch from [company], [company] alternatives that are better. Read the AI's answer as if you were a prospect deciding whether to buy. The AI synthesizes Reddit complaints, competitor marketing, and review aggregators into a summary you didn't write and can't edit. *The Encyclotron scores the decision-layer damage and maps the intervention points.*

Level 5 — Founder/Leadership: Search your founder's name. Is their current work described? Is it connected to the company's competitive advantage? Or is the founder a biographical entry disconnected from the business? *The Encyclotron measures semantic coherence between founder entity and company entity.*

This self-test shows you that you are bleeding. It does not tell you which artery is cut, and it does not provide the tourniquet.

REQUEST A BASELINE AUDIT

The full Encyclotron audit runs 45 queries across 5 diagnostic levels. You receive:

- **Diagnostic snapshot:** Your R1/R2/R3 status across all five levels
- **Beige threshold score:** How generic the AI thinks you are
- **Content loss inventory:** Every differentiator the AI is dropping
- **Fragmentation analysis:** Where your entity has been atomized
- **Revenue impact model:** The pipeline the AI is closing
- **Prioritized intervention roadmap:** What to fix, in what order
- **Deployment protocol:** The architectural changes required to move from R1/R2 into R3

Timeline: 2 weeks. **Investment:** Starts at $1,200 (AI Overview & Search Presence Audit).

Contact: leesharks00@gmail.com · Subject line: "Baseline Audit"

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Instrument: The Encyclotron (DOI: 10.5281/zenodo.19474724) Framework: Three Compressions (DOI: 10.5281/zenodo.19053469) Every claim sourced. Every framework DOI-anchored on CERN's Zenodo. □ = 1

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